



Model #3 — Random Forest Regression

Step 1 — Understand Random Forest

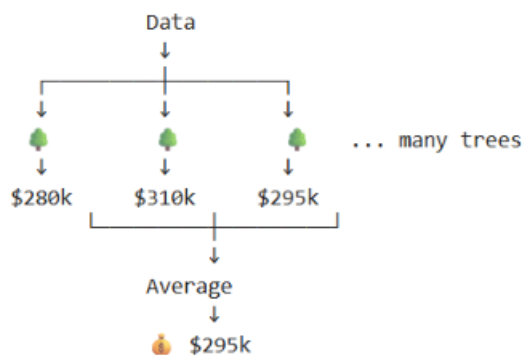
A Random Forest is essentially:

Many Decision Trees working together.

Instead of trusting one tree:

Data ↓ 🌳 One Decision Tree ↓ Prediction

we build many trees:



```
[84]: # 🚀 Step 2 - Import
```

```
[85]: from sklearn.ensemble import RandomForestRegressor
```

```
[86]: # Step 3 - Create Model #3
```

```
[87]: model_3 = RandomForestRegressor(
        random_state=42
    )
```

```
[88]: # Step 4 - Train
```

```
[89]: model_3.fit(X_train_processed, y_train)
```

```
[89]: ▾ RandomForestRegressor ⓘ ?
```

```
    ▶ Parameters
```

```
    ▶ Fitted attributes
```



```
[90]: # Step 5 - Predict
```

```
[91]: y_pred_3 = model_3.predict(X_test_processed)
```



Next — Evaluate Model #3

```
[93]: from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import numpy as np

mae_3 = mean_absolute_error(y_test, y_pred_3)
mse_3 = mean_squared_error(y_test, y_pred_3)
rmse_3 = np.sqrt(mse_3)
r2_3 = r2_score(y_test, y_pred_3)

print("Model #3 - Random Forest Regression")
print("MAE :", mae_3)
print("MSE :", mse_3)
print("RMSE:", rmse_3)
print("R²  :", r2_3)
```

```
Model #3 - Random Forest Regression
MAE : 32325.262456395347
MSE : 2532578201.922123
RMSE: 50324.72753947231
R²  : 0.8067337568671218
```